

Prathamesh Shete

AI/ML Engineer | Generative AI & Transformer Specialist

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 Pune, Maharashtra

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Professional Summary

AI Engineer with 2.6+ years of experience designing, developing, and deploying LLM-based and Transformer-driven systems in production. Proven expertise in Generative AI, deep learning, and data-centric model development using Python, TensorFlow, Hugging Face, and modern RAG frameworks. Skilled in building scalable ML pipelines, fine-tuning language models, and optimizing retrieval systems for real-world business applications. Strong background in AWS deployment, model evaluation, and MLOps-oriented automation for production-grade AI systems.

Core Competencies

Generative AI & LLMs

- Transformer Architectures
- Model Fine-tuning
- Text Generation
- Embeddings
- RAG Systems

Machine Learning

- Model Training & Evaluation
- Feature Engineering
- Deep Learning (CNNs, RNNs, LSTMs)

Programming & Frameworks

- Python
- FastAPI
- LangChain
- TensorFlow
- Hugging Face
- Scikit-learn

MLOps & Deployment

- Docker
- Redis
- REST APIs
- AWS (EC2, S3, Lambda)
- GPU Utilization

Tools & Libraries

- FAISS
 - Chroma
 - LlamaIndex
 - OpenAI SDK
 - Weights & Biases
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Professional Experience

AI Engineer

Rupadi International Pvt Ltd | Mar 2025 – Present | Pune

- Designed and deployed an LLM-driven interviewer system using LangChain, Redis, and Python for adaptive conversational evaluation.
- Built Transformer-based pipelines for real-time question generation and semantic analysis.
- Developed modular FastAPI microservices enabling scalable LLM inference and seamless REST integration.
- Utilized AWS GPU instances to optimize model inference speed and reduce latency.
- Collaborated closely with backend and frontend teams to integrate AI workflows into production systems.

qData Scientist (Junior)

Teamlease | Aug 2024 – Dec 2024 | Viman Nagar, Pune

- Built RAG pipelines using Hugging Face Transformer models to improve factual grounding and retrieval quality.
- Engineered data preprocessing workflows for large-scale text and unstructured datasets.
- Performed evaluation and fine-tuning for generative tasks using BLEU, ROUGE, and accuracy metrics.
- Contributed to containerization and reproducible experiment setups using Docker.

Data Researcher

Collegedunia | Mar 2023 – May 2024 | Pune

- Automated data extraction using BeautifulSoup and Selenium, generating structured knowledge representations.
 - Supported the development of a Generative chatbot using FAISS and RAG for document-based question answering.
 - Ensured high-quality data labeling and source validation to maintain dataset integrity.
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Key Projects

AI Interviewer App

- Built a multimodal generative interviewer chatbot integrating LLMs, speech-to-text processing, and FastAPI microservices.
- Deployed scalable AI workflows on AWS infrastructure.

Admission Query Chatbot

- Implemented a Transformer-powered education chatbot leveraging FAISS-based semantic retrieval for improved query resolution.

AI Calling Agent – Poll Pandit

- Built an AI-powered outbound telecalling system for Marathi-language farmer surveys using autonomous voice conversations.
- Integrated Google Gemini Live for real-time conversational AI and natural voice synthesis.
- Used Vobiz/Twilio/Exotel APIs for automated phone call handling and campaign execution.

- Implemented Groq LLM pipelines for answer extraction, sentiment analysis, and structured reporting.
 - Designed backend services using Python, FastAPI, and MongoDB for scalable survey tracking.
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Education

B.E., Electrical Engineering

SB Patil College, Indapur (2017 – 2020)

Diploma, Electrical Engineering

KBP Polytechnic, Satara (2014 – 2017)

Certifications

- Generative AI – Databricks
 - LangChain for LLM Apps – Udemy
 - Full Stack Data Scientist – iNeuron
 - SQL & Power BI – Codebasics
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Soft Skills

- Analytical Thinking
- Attention to Detail
- Clear Communication
- Collaborative Mindset
- Research-Oriented
- Adaptable Learner